



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 25 Review for Final Exam

- Last quiz and guest lecture: May 7th, 5:30-7:00pm, Location TBA
- Final Exam: May 13rd, 8:30-10:30am, Location TBA
- Zero tolerance policy when it comes to cheating so, **DON'T CHEAT**
- Missing the final exam without prior notification to and approval of the instructors will automatically result in the "0" grade for the exam.
- You can bring one calculator, one non-electronic dictionary, **no cheat sheet.**

How to prepare for the final exam?

- The types of questions on the final exam will be very similar to **last year's final exam**, which will be released soon for your reference.
- It is highly recommended that you carefully review last year's questions and solutions to familiarize yourself with the **exam format** and **key test points**.
- Feel free to schedule an (online) office hour with me if you need any help in preparing for the final exam. My zoom link:

<https://cuhk-edu-cn.zoom.us/j/5533045922>

Probability

Probability Terminologies

- **Random Experiment:** a repeatable procedure
- **Sample space:** set of all possible outcomes Ω .
- **Event:** a subset of the sample space.
- **Probability mass function (discrete), $P(\omega)$:** gives the probability for each outcome $\omega \in \Omega$
- **Probability density function (continuous), $f(x)$:**

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Probability Distributions

- Bernoulli Distribution: take value 1 with probability p and value 0 with probability $1 - p$. Mean = p . Variance = $p(1 - p)$.
- Binomial Distribution: the number of success in N independent experiments (for each experiment, the probability of success is p). Mean = Np . Variance = $Np(1 - p)$.
- Geometric Distribution: the number of experiments needed to get the first success (for each experiment, the probability of success is p). Mean = $1/p$. Variance = $(1 - p)/p^2$.

Probability Distributions

- Uniform Distribution: with the same likelihood, X takes a value within $[a, b]$ where $b > a$. $f(x) = 1/(b - a)$ for $x \in [a, b]$, and $f(x) = 0$ otherwise. Mean = $(a + b)/2$. Variance = $(b - a)^2/12$.
- Normal Distribution: Mean = μ . Variance = σ^2 .

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

Cumulative Distribution Function

- The CDF of a discrete random variable X is

$$F(x) = P(X \leq x) = \sum_{\tilde{x} \leq x} f(\tilde{x})$$

- CDF for continuous random variable is defined as:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du$$

✓ $0 \leq F(x) \leq 1$

✓ If $x \leq y$, then $F(x) \leq F(y)$

For both discrete and continuous RVs

Mean and Variance

- Discrete:
 - ✓ Probability mass function.
- Continuous
 - ✓ Probability density function.

Summation $\leftrightarrow$ Integration

- Mean

$$E[X] = \sum x f(x)$$

- Variance

$$\text{Var}[X] = \sum (x - E[X])^2 f(x)$$

- Mean

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

- Variance

$$\text{Var}[X] = \int_{-\infty}^{\infty} (x - E(X))^2 f(x) dx$$

Expectation

- $E[\sum_i C_i X_i] = \sum_i C_i E[X_i]$
- $E[C] = C$
- $E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$

Conditional Probability

- Given the realization of event A, the probability of event B may change
- (**Conditional probability**) If A and B are events with $P(B) > 0$, then the conditional probability of A given B, denoted by $P(A|B)$, is defined as
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
- (**Independence**) Two events A and B are called independent if and only if $P(A \cap B) = P(A)P(B)$
$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

Statistics

Maximum Likelihood Estimate

- Given a model with an unknown parameter θ
- Given samples: $X_1, X_2 \dots, X_n$
- The probability that the models generates the samples is called **likelihood**. $L(\theta) = P(X_1, X_2 \dots, X_n | \theta)$
- To determine the best $\theta \in \Theta$, we choose $\hat{\theta}$ such that $L(\theta)$ is maximized at $\theta = \hat{\theta}$. **Maximum likelihood estimate (MLE)**

Likelihood Function

- Given a model with an unknown parameter θ
- Given samples: $X_1, X_2 \dots, X_n$

Continuous RV model:

- Likelihood: $L(\theta) = \prod_i f(X_i | \theta)$
- Log-Likelihood: $l(\theta) = \sum_i \log(f(X_i | \theta))$

f : the model's probability density function (PDF)

Discrete RV model:

- Likelihood: $L(\theta) = \prod_i P(X_i | \theta)$
- Log-Likelihood: $l(\theta) = \sum_i \log(P(X_i | \theta))$

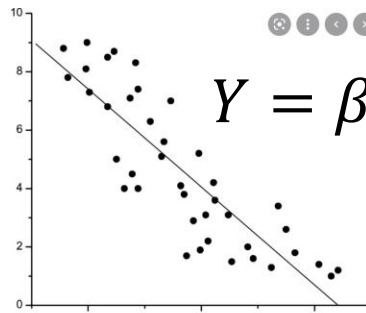
P : the model's probability mass function (PMF)

Example

- Suppose heights of students follow a normal distribution.
 - Given parameter μ , the model is $N(\mu, 1)$
- Samples: $X_1, X_2, \dots, X_n$
- $L(\mu) = \left(\frac{1}{\sqrt{2\pi}}\right)^n e^{-\frac{1}{2}\sum_i (X_i - \mu)^2}$
- $l(\mu) = \text{Log } L(\mu) = -\frac{1}{2}\sum_i (X_i - \mu)^2 - \frac{n}{2}\log 2\pi$
- $l'(\mu) = \sum_i (X_i - \mu) = 0 \quad \longrightarrow \quad \hat{\mu} = \bar{X}.$

Hint: Go through all homework problems in Hw 2 (Lecture 9)

Linear Regression



$$Y = \beta_0 + \beta_1 X$$

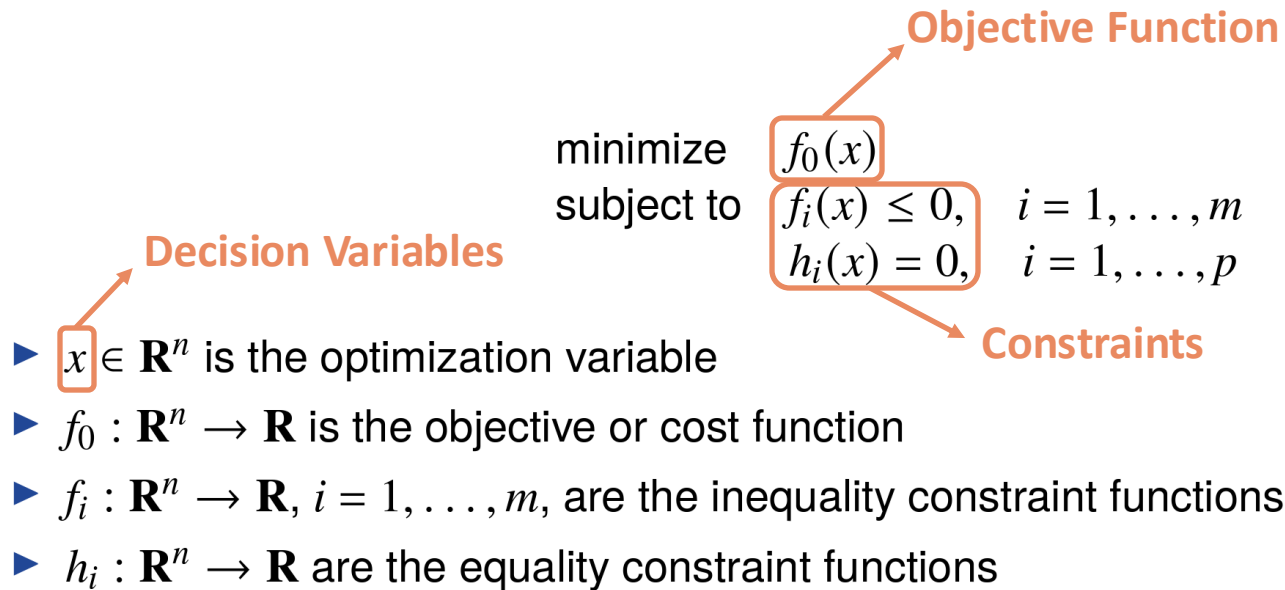
- $Y \sim N(\beta_0 + \beta_1 X, \sigma^2)$
 - Given the observation of X
 - Y follows a normal distribution with mean $\beta_0 + \beta_1 X$
- Regression analysis: knowing $\beta_0, \beta_1, \sigma^2$, you can predict X given Y

MLE: choose the best β_0, β_1

$$\text{MLE: } \left\{ \begin{array}{l} \widehat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} \\ \widehat{\beta}_0 = \bar{Y} - \widehat{\beta}_1 \bar{X} \end{array} \right.$$

Optimization

Optimization problem in standard form

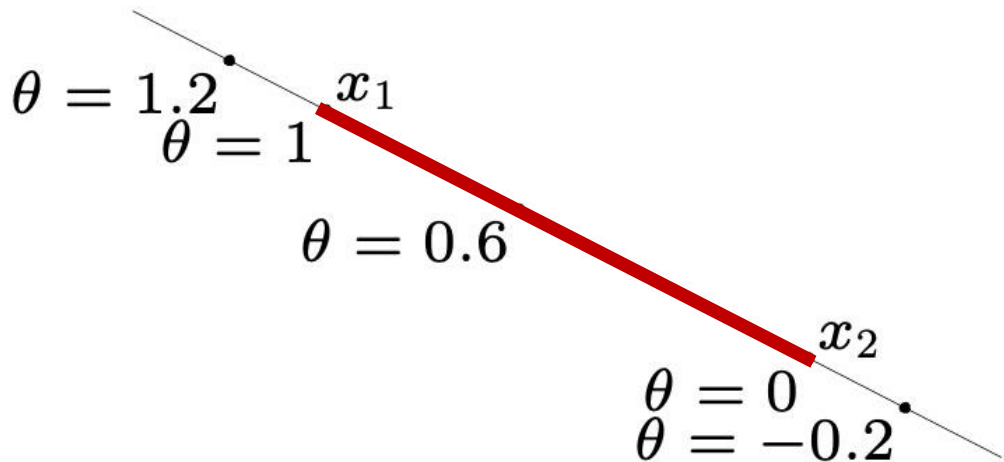


Line Segment

- Let $x_1 \neq x_2$ be two points in $\mathbb{R}^n$. Points of the form

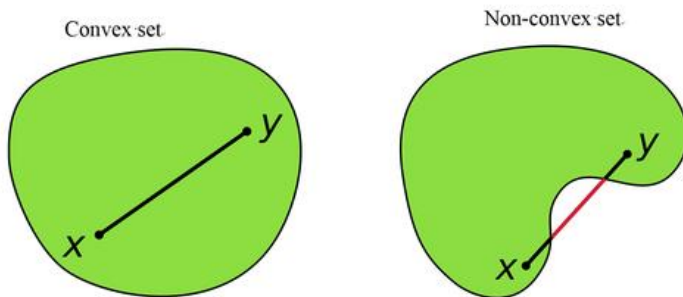
$$x = \theta x_1 + (1 - \theta)x_2$$

where $\theta \in [0, 1]$, form the **line segment** between x_1 and x_2 .



Convex Set

- Set C is a **convex set** if the line segment between any two points in C lies in C .



- Formal definition: A set C is convex if $\forall \mathbf{x}_1, \mathbf{x}_2 \in C, \forall \theta \in [0,1]$
 $\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2 \in C$.

Remark: In this lecture, I will use **bold** form to represent a high dimension point. Without bold form, it represents a scalar

Convex Set Examples

- The empty set $\emptyset$, the singleton set $\{\mathbf{x}_0\}$, and the complete space R are convex sets.
- An interval of $[a, b] \subset R$ is a convex set
- In R^n the set $H := \{\mathbf{x} \in R^n : a_1x_1 + \dots + a_nx_n = c\}$ is a convex set
- Half spaces, e.g., $H := \{(x, y) : y \leq ax + b\}$ are convex sets
- A disk with center $(0,0)$ and radius c is a convex subset of R^2

Remark: In this lecture, I will use **bold** form to represent a high dimension point. Without bold form, it represents a scalar

Steps for Showing the Convexity of a Set

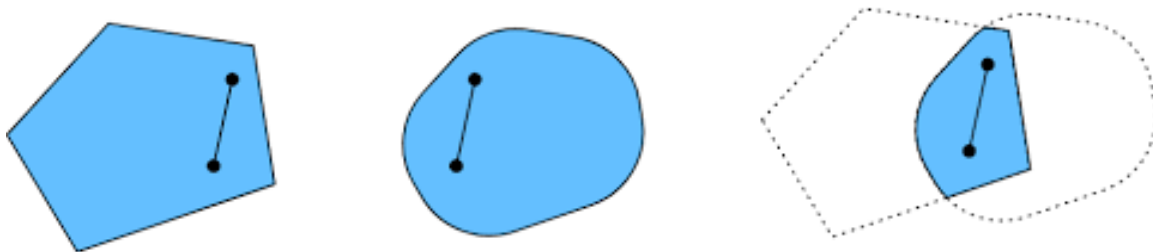
Prove $H := \{(x, y) : y = ax + b\}$ is a convex set

For any (x_1, y_1) and (x_2, y_2) in H ,

- $y_1 = ax_1 + b$
- $y_2 = ax_2 + b$
- $\theta(x_1, y_1) + (1 - \theta)(x_2, y_2) = (\underbrace{\theta x_1 + (1 - \theta)x_2}_{\text{1. Use the assumption that } (x_1, y_1), (x_2, y_2) \in H}, \underbrace{\theta y_1 + (1 - \theta)y_2}_{\text{2. Characterize the new point within the line segment}})$
- Then for any $\theta \in [0, 1]$
 - $\underbrace{\theta y_1 + (1 - \theta)y_2}_{\text{3. Use (1) and (2) to show that the new point is in } H} = a(\theta x_1 + (1 - \theta)x_2) + b$

Property of Convex Sets

Lemma: If both S_1 and S_2 are convex sets, then $S_1 \cap S_2$ is also a convex set.



Hint: Go through homework problems
1.6, 2.2, 2.3, 2.7,.2.8 and similar
problems in Hw 3

Convex Function

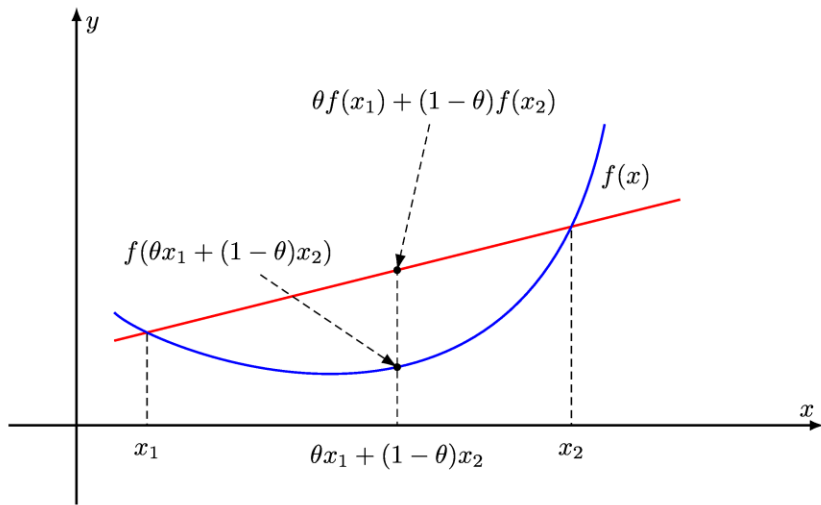
Definition: A function $f(x): R^n \rightarrow R$ is **convex** if (1) its domain is a convex set, and

(2) for any $x_1, x_2 \in \text{dom}(f)$ and any $0 \leq \lambda \leq 1$, we have

$$f(z) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

where $z = \lambda x_1 + (1 - \lambda)x_2$.

Function f evaluated at the combination of two points x_1, x_2 is **no larger than** the same combination of $f(x_1)$ and $f(x_2)$

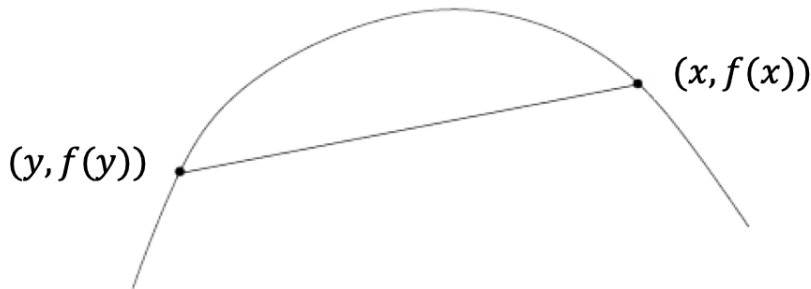


Concave Function

Definition: A function $f(x): R^n \rightarrow R$ is **concave** if (1) the domain of f is a convex set, and (2) for any $x, y \in \text{dom}(f)$ and any $0 \leq \lambda \leq 1$, we have

$$f(z) \geq \lambda f(x) + (1 - \lambda)f(y)$$

where $z = \lambda x + (1 - \lambda)y$.



If f is concave, then $-f$ is convex!

If f is convex, then $-f$ is concave!

Second Order Condition

Suppose f is a **twice continuously differentiable** function. Then f is convex **if and only if**

(1) $\text{dom}(f)$ is a convex set

(2) for any $\mathbf{x} \in \text{dom}(f)$, any vector $\mathbf{e}$ satisfying that there exists $t > 0$ such that $\mathbf{x} + t\mathbf{e} \in \text{dom}(f)$,

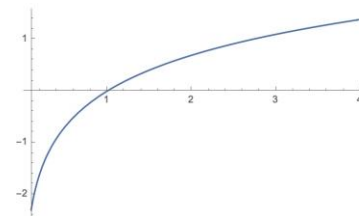
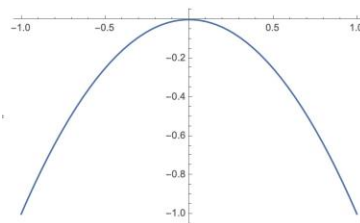
$$\frac{d^2 f(\mathbf{x} + t\mathbf{e})}{dt^2} (0) \geq 0$$

One dimension: **$f''(\mathbf{x}) \geq 0$**

Examples

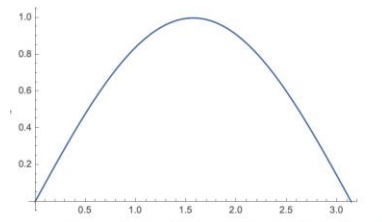
Convex

- $f(x) = ax + b$ (also concave)
- $f(x) = x^2$
- $f(x) = e^x$



Concave

- $f(x) = -x^2$
- $f(x) = \log(x)$ on $(0, +\infty)$
- $f(x) = \sin(x)$ on $[0, \pi]$



Convex Function VS. Convex Set

- $C = \{\mathbf{x}: f(\mathbf{x}) \leq r\}$ is a convex set if $f(\mathbf{x})$ is a convex function
 - C is also called a **sublevel set** of $f(\mathbf{x})$
- $C = \{(\mathbf{x}, y): y \geq f(\mathbf{x})\}$ is a convex set **if and only if** $f(\mathbf{x})$ is a convex function.
 - C is also called the **epigraph** of $f(\mathbf{x})$

Application

Prove a unit disk, e.g., $H := \{(x, y): x^2 + y^2 \leq 1\}$ is a convex set.

We consider $f(\mathbf{x}) = \sum_i a_i x_i^2$ with $a_i > 0$. Given any point $\mathbf{y}$, any unit vector $\mathbf{e}$ and any θ

$$g(\theta) = f(\mathbf{y} + \theta \mathbf{e}) = \sum_i a_i (y_i + \theta e_i)^2$$

Then $g''(0) = \sum_i 2a_i e_i^2 \geq 0$. So $f(\mathbf{x})$ is a convex function.

$\{\mathbf{x}: f(\mathbf{x}) \leq r\}$ forms a ball/disk or an ellipsoid, so it is a convex set.

Operations Preserving Convexity

- If f is convex, then f/c is convex for any $c > 0$
- If f_1, f_2 are convex, then $f_1 + f_2$ are convex
- **Affine Transformation:** If f is convex, then $g(\mathbf{x}) = f(a\mathbf{x} + \mathbf{b}) - c$ is also convex

- **Pointwise Maximum:** If $f_1, \dots, f_m$ are convex, then the function

$$f(\mathbf{x}) = \max\{f_1(\mathbf{x}), \dots, f_m(\mathbf{x})\}$$

is also convex.

- **Non-negative Weighted Sums:** If $f_1, f_2, \dots, f_n$ are convex functions, and $w_1, w_2, \dots, w_n \geq 0$, then the function

$$f(x) = w_1 f_1(x) + w_2 f_2(x) + \dots + w_n f_n(x)$$

is convex.

Hint: Go through homework problems
3.1, 3.2, 3.3, 3.6, 3.8, 5.1 and similar
problems in Hw 3

Convex Optimization Problem

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & g_i(x) \leq 0, i = 1, \dots, m \\ & h_i(x) = 0, i = 1, \dots, n\end{array}$$

A **convex optimization problem** needs to satisfy the following two conditions:

- Its feasible set is a **convex set**.
- Its objective function is a **convex function**.

Why Convex Optimization Problem?

- Any local minimum is also a global minimum.
- Any interior local minimum satisfies the first order condition.

$$\nabla f(p) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(p) \\ \vdots \\ \frac{\partial f}{\partial x_n}(p) \end{bmatrix} \xrightarrow{\quad} \nabla f(\mathbf{x}^*) = \mathbf{0}$$

Example: Newsvendor Problem

- Suppose you want to start your own blind box business.
- Let D denote the one season (three months) random demand, which follows a **uniform distribution in $[10,100]$** .
- At the beginning of each season, you place an order Q to Pop Mart, with a cost **10 Yuan** for each blind box.
- Each blind box can be sold at a price of **20 Yuan**.
- At the end of each season, unsold blind boxes are salvaged, and you get **3 Yuan** for each salvaged box.
- How many blind boxes should you order to maximize your expected profit?— You should know how to find the optimal solution!

Review: Newsvendor Problem

- A newsvendor must decide how many newspapers to purchase each morning to maximize expected profit.
 - Demand D is random, with probability density function $f(\cdot)$ and cumulative distribution function $F(\cdot)$.
 - Costs & Revenues
 - Purchase cost per newspaper: c
 - Selling price per newspaper: p
 - Salvage value: 0 (unsold newspapers have no residual value)
 - Decision variable: Order quantity q



Review: Newsvendor Problem

- Objective

- Sales: $\min(q, D)$
- Expected revenue: $E_D[p \cdot \min(q, D)]$
- Ordering cost: cq
- Expected profit

$$\pi(q) = E_D[p \min(q, D)] - cq$$

- *Exercise: Show π is concave, or equivalently, $-\pi$ is convex.*
 - Hint: using operations that preserve convexity



Review: Newsvendor Problem

Proof. (of convexity of $-\pi(q) = -E_D[p \min(q, D)] + cq$)

1. The function $g(q) = -\min(q, D)$ is convex in q
2. The function $h(g(q)) = pg(q)$ is convex in q
3. The expectation $-E_D[p \min(q, D)]$ is convex in q
4. The objective $-\pi(q)$ is convex

Review: Newsvendor Problem

Proof. (of convexity of $-\pi(q) = -E_D[p \min(q, D)] + cq$)

1. The function $g(q) = -\min(q, D)$ is convex in q
 - Because $g(q) = -\min(q, D) = \max(-q, -D)$ is the maximum of linear functions of q
2. The function $h(g(q)) = pg(q)$ is convex in q
 - Because scaling by a positive constant $p > 0$ preserves convexity
3. The expectation $E_D[p \min(q, D)]$ is convex in q
 - Because weighted sum of convex functions is still convex
4. The objective $-\pi(q)$ is convex
 - Because cq is linear (thus convex), the sum preserves convexity

Review: Newsvendor Problem: Finding the Optimal Order Quantity

- Express the objective as an integral

$$\pi(q) = p \int_0^q x f(x) dx + pq(1 - F(q)) - cq.$$

- To maximize $\pi(q)$ take the derivative with respect to q :

$$\frac{d}{dq}\pi(q) = p \cdot qf(q) + p(1 - F(q)) - pqf(q) - c$$

- Simplifies to:
$$\frac{d}{dq}\pi(q) = p(1 - F(q)) - c$$

Review: Newsvendor Problem: Finding the Optimal Order Quantity

- Setting $\frac{\partial}{\partial q} \pi(q) = 0$:

$$F(q^*) = \frac{p - c}{p}.$$

- Solving for q^* :

$$q^* = F^{-1} \left(\frac{p - c}{p} \right)$$

 The optimal order quantity is directly linked to demand uncertainty and price-cost ratio.

 This reinforces why convex optimization is a powerful tool in real-world decision-making! 

Machine Learning

Supervised Learning

- **Supervised machine learning** algorithms utilize **labeled data** for training, where the correct outputs corresponding to input data are already known.
- For all samples, (x^i, y^i) , $i = 1, \dots, N$, you can observe both the input data x^i and the label y^i

Training data



$y=1$ (cat)



$y=0$ (dog)



$y=1$ (cat)

...



$y=0$ (dog)

Supervised Learning

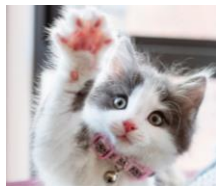
Training data



$y=1$ (cat)



$y=0$ (dog)

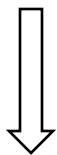


$y=1$ (cat)

...



$y=0$ (dog)



Learning algorithm (optimization involved)

Classifier $h: X \rightarrow \{0,1\}$

For example:

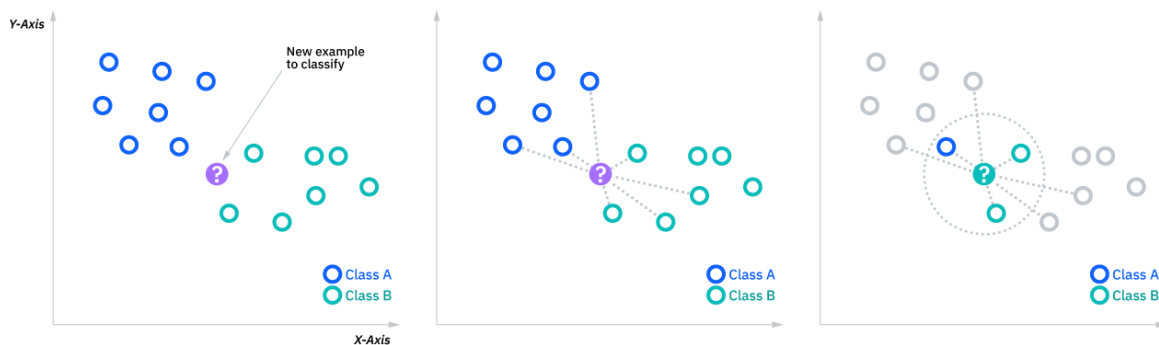
$h \left(\text{image of a dog} \right)$



0

K-Nearest Neighbor Classifier

- Find K training points x_i closest to x .
- If the **majority** of K -nearest neighbors of x belong to classifier c , label x as c .

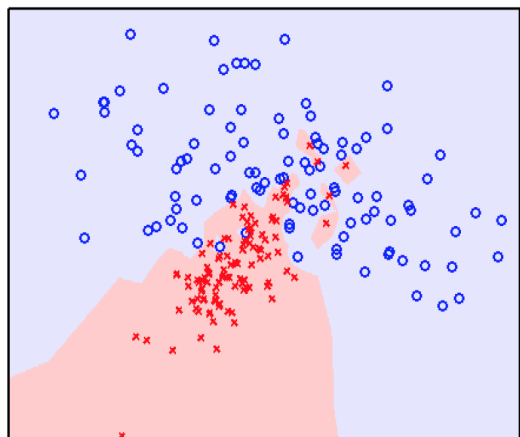


The KNN Algorithm

1. Load the data
2. Set K of your choice to be the number of neighbors
3. For each new data to be classified
 - Calculate the distances between the new data and all the labeled data.
 - Record the entry (d_i, y_i) , where d_i is the distance between the new data and the i th labeled data, and y_i is the label of the i th data.
 - Sort these entries with respect to distance (from smallest to largest).
5. Pick the first K entries from the sorted collection
6. Get the labels of the selected K entries
7. Choose the label with the largest frequency

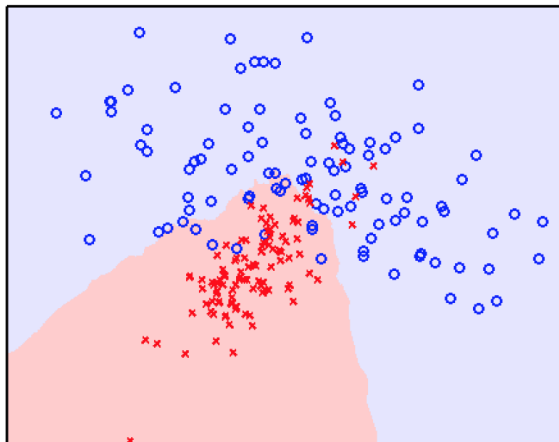
The Choice of K

Overfitting

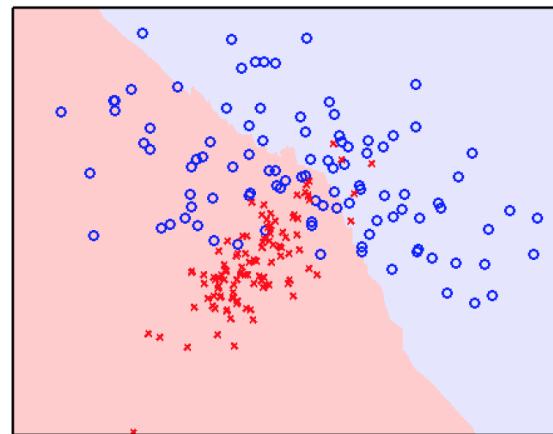


$K = 1$

Underfitting



$K = 25$



$K = 101$

Overfitting and Underfitting

- A machine learning model is said to **overfit** the data when it learns patterns specific to the training data and make accurate predictions only on the training data.
- A machine learning model is said to **underfit** when it fails to capture the key patterns or relationships between variables in both the training and test data.

Logistic Regression

- Model the conditional probability of the label given the data
- **Simplest** case (two classes): $y \in \{0, 1\}$
- **Logistic regression model:**

Logistic function

$$p(y = 1|\mathbf{x}, \boldsymbol{\theta}, b) = \frac{1}{1 + \exp(-(\boldsymbol{\theta}^\top \mathbf{x} + b))}$$

$$p(y = 0|\mathbf{x}, \boldsymbol{\theta}, b) = \frac{\exp(-(\boldsymbol{\theta}^\top \mathbf{x} + b))}{1 + \exp(-(\boldsymbol{\theta}^\top \mathbf{x} + b))}$$

Train the Logistic Regression Model

- How to find θ and b ? MLE

- Given m labeled samples $(\mathbf{x}^i, y^i)$, $i = 1, \dots, m$
- Find θ and b such that the likelihood of observing the labeled samples is maximized

$$\max_{\theta, b} l(\theta, b) := \log \prod_{i=1}^m P(y^i | \mathbf{x}^i, \theta, b) = \sum_{i=1}^m \log P(y^i | \mathbf{x}^i, \theta, b)$$

- Usually, we equivalently maximize the averaged likelihood

$$\max_{\theta, b} \frac{1}{m} l(\theta, b) := \frac{1}{m} \sum_{i=1}^m \log P(y^i | \mathbf{x}^i, \theta, b)$$

Bad news: no
closed form
solution to the
problem

Gradient Descent Method

- Start with an initial point $x^{(0)}$

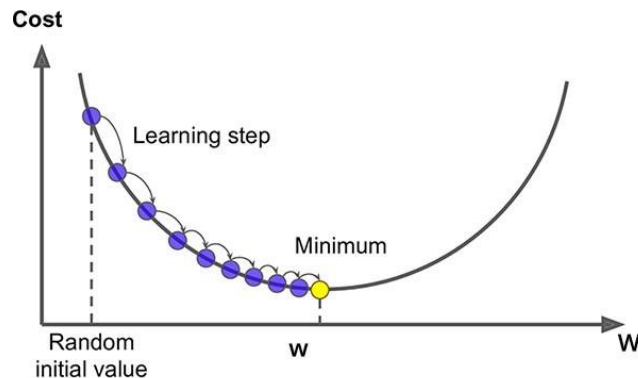
$\alpha^{(t)}$: the step size or learning rate

- Update our point by the following rule:

$$x^{(t+1)} = x^{(t)} - \boxed{\alpha^{(t)}} f'(x^{(t)})$$

- Stopping criteria:

- $|x^{(t+1)} - x^{(t)}| \leq \varepsilon$
- or $|f'(x^{(t)})| \leq \varepsilon$



How to select $\alpha^{(t)}$? The selection of $\alpha^{(t)}$ will affect the rate at which we find the local minimizer. A bad selection of $\alpha^{(t)}$ can result in the failure of the algorithm.

If $\alpha^{(t)}$ is not well chosen, we may not meet the stop criteria.

$$f(x) = x^2$$

Suppose $\alpha^{(t)} = 1$.

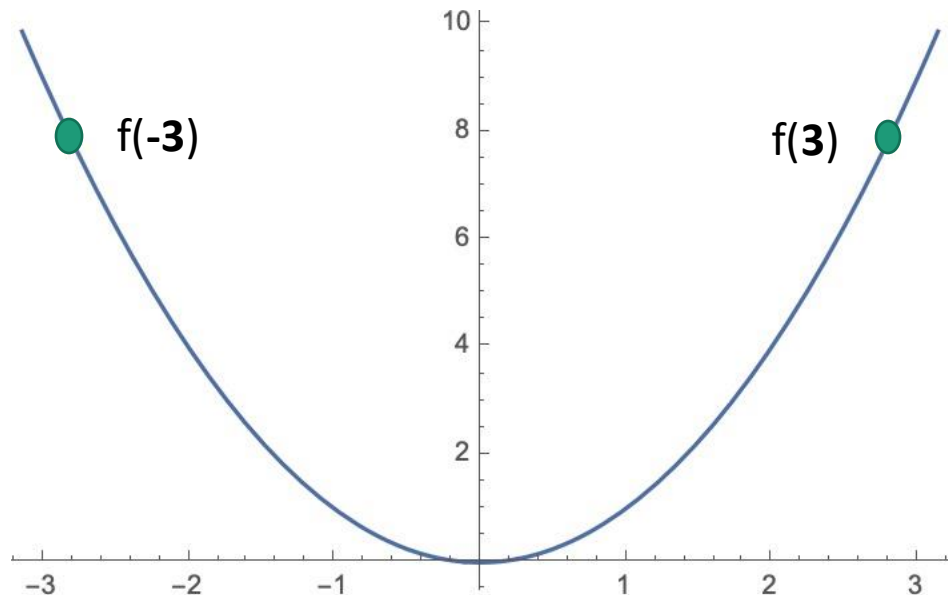
$$x^{(t+1)} = x^{(t)} - f'(x^{(t)})$$

If $x^{(t)} = -3$

- $f'(-3) = -6$
- $x^{(t+1)} = 3$

If $x^{(t)} = 3$

- $f'(3) = 6$
- $x^{(t+1)} = -3$



The updated points will oscillate between 3 and -3

Unsupervised Learning

- Data lacks structured or objective answers, such as labels.
- In other words, for all samples (x^i, y^i) , where $i = 1, \dots, N$, you can observe x^i but y^i remains unseen.

Training data



~~$y=1$ (cat)~~



~~$y=0$ (dog)~~



~~$y=1$ (cat)~~

...



~~$y=0$ (dog)~~

Unsupervised Learning

- There is no predefined correct output for a given input.
- Instead, the algorithm must interpret the input and make the appropriate decision.
- The aim is to **examine the data and discern underlying patterns.**

K-Means Clustering

- Given m data points, $\{x^1, x^2, \dots, x^m\}$
- Find k cluster centers, $\{c^1, c^2, \dots, c^k\}$
- And assign each data point i to one cluster, $\pi(i) \in \{1, \dots, k\}$
- Such that the sum of the distances from each data point to its respective cluster center is minimized

K-Means Clustering

- Step 1: Initialize k cluster centers, $\{c^1, c^2, \dots, c^k\}$, randomly
- Step 2: Do
 - Decide the cluster memberships of each data point, x^i , by assigning it to the nearest cluster center (**cluster assignment**)

$$\pi(i) = \underset{j=1, \dots, k}{\operatorname{argmin}} \|x^i - c^j\|^2$$

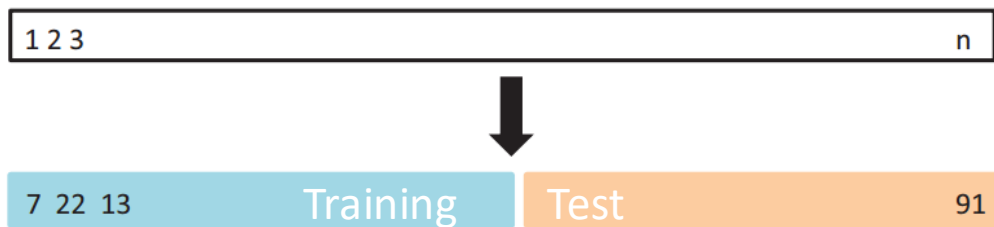
- Adjust the cluster centers (**center adjustment**)

$$c^j = \frac{1}{|\{i: \pi(i) = j\}|} \sum_{i: \pi(i)=j} x^i$$

- While any cluster center undergoes changes, go to Step 2

Model Selection: Validation Set Approach

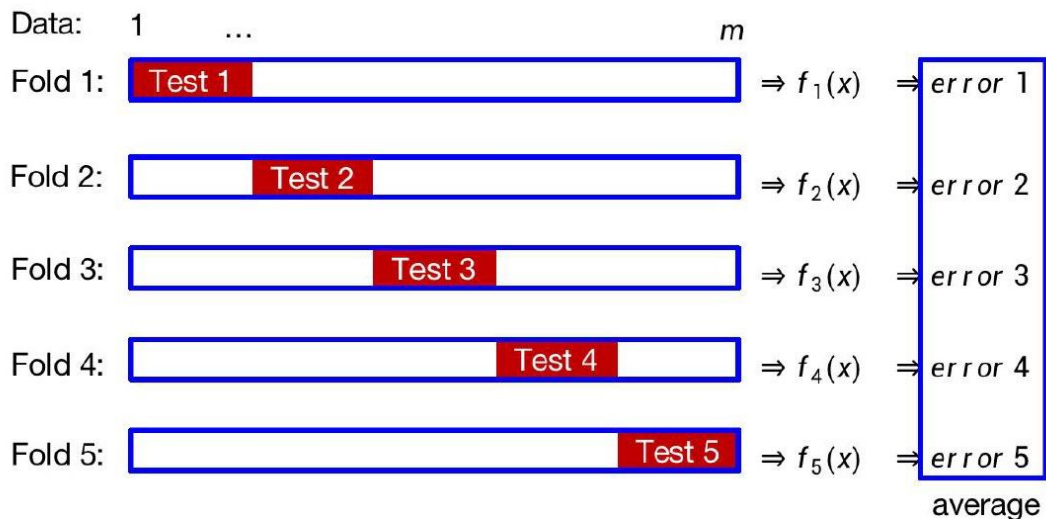
- Divide samples in to **training data** and **test data**.



- A set of model to choose $\{1, 2, \dots, M\}$. (e.g, KNN, choose $K=1, 2, \dots, M$)
- For each model m ,
 - use training data to train model
 - use the learned model to calculate the prediction error for test data. Err_m
- Choose model that has the smallest test error ($\min \text{Err}_m$)

K-Fold Validation

- 5-fold cross-validation (blank: training; red: test)



- f_i is fitted by the training data in Fold i .
- For each fold, use test data to test the prediction error.
- Use the **average** error as the model's prediction error.

Error

Model 1 $h(\theta, X)$	Model 2 $g(\gamma, X)$
$Err_h(\theta_1)$	$Err_g(\gamma_1)$
$Err_h(\theta_2)$	$Err_g(\gamma_2)$
$Err_h(\theta_3)$	$Err_g(\gamma_3)$
$Err_h(\theta_4)$	$Err_g(\gamma_4)$
$Err_h(\theta_5)$	$Err_g(\gamma_5)$

Model 1 is better iff.

$$\frac{1}{K} \sum_i Err_h(\theta_i) < \frac{1}{K} \sum_i Err_g(\gamma_i)$$

Hint: Go through homework problems
2.6, 2.7, 2.10, 3.3, 3.4 and similar
problems in Hw 4

Best of luck on your final exam! I hope you found this course valuable and took away something meaningful. 😊